Nisumba Soodhani K.

Ph.D. Candidate, Educational Technology
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Research Profile

I study persistence in makerspaces and technology-enhanced learning environments. I use interaction analysis and mixed-method approaches to examine how learners navigate challenges during collaborative STEM activities. My research contributes to the design of learning environments that support meaningful learning. My broader interests include learning analytics and AI in education.

Education

Ph.D. in Educational Technology (Thesis submission - Nov 2026) **2021 – Present**
IIT Bombay *Mumbai, India*

- **Thesis:** Characterizing Persistence as a Process in Collaborative Electronic Making: An Interactional Account
- **Supervisors:** Prof. Ramkumar Rajendran and Prof. Sridhar Iyer **CPI:** 9.79/10

M.E. in Power Electronics **2017 – 2019**
College of Engineering Guindy, Anna University *Chennai, India*

- **Thesis:** An External Rotor Synchronous Reluctance Motor Analysis For Electric Assisted Bike
- **Supervisor:** Dr. Kavitha A **CPI:** 8.10/10

B.Tech in Electrical and Electronics Engineering **2013 – 2017**
SASTRA University *Thanjavur, India*

- **Thesis:** Investigation on non isolated high gain convertor
- **Supervisor:** Dr. Girish Ganesan R **CPI:** 7.82/10

Research Experience

Research Scholar, Learning Analytics Lab **2023 – Present**
IIT Bombay *Mumbai, India*

- Temporal dynamics of team's verbal interactions during Collaborative Problem Solving (CPS) in classroom settings
- Affective computing in collaborative problem solving settings.

Junior Research Fellow **2020 – 2021**
National Institute of Technology, Tiruchirappalli *Tiruchirappalli, India*

- MEMS — Wireless Sensor Node for online data transfer of parameters from electrical machines and drives.
- Supervisor: Dr. S Moorthi

Projects

1. Doctoral Research: Motivation and Persistence in Collaborative Making **2022 – Present**
Supervisors: Prof. Ramkumar Rajendran and Prof. Sridhar Iyer *IIT Bombay*

- Investigating motivation and persistence in collaborative electronic making through video-based interaction analysis.
- Examining how learners navigate challenges during STEM design activities and collaborative problem solving.
- Using process analysis and qualitative analysis to understand behavioural manifestations of persistence (productive persistence and wheel spinning).
- Identified the interactional processes through which teams sustain productive persistence during extended making activities.

2. Pi Jam and Maker Factory Ethnography **2023 – 2024**
Supervisor: Prof. Ramkumar Rajendran *IIT Bombay*

- Conducted three-month fieldwork across BMC (Mumbai) government schools to study the implementation of low-cost makerspace programmes (Pi Jam and Maker Factory) for middle school learners.
- Conducted interviews with facilitators and classroom observations; analysed data using thematic analysis to examine pedagogical practices, student engagement, gender inclusivity, and maker culture.

3. Critical Action Learning Exchange (CALE) **2022 – 2023**
Supervisor: Prof. Chandan Dasgupta *IIT Bombay*
PI: Prof. Jim Slotta *University of Toronto*

- Contributed to an international research initiative engaging teachers across different countries in designing culturally responsive, action-oriented lessons addressing social and environmental challenges.
- Collected and analysed data from Indian teachers participating in CALE.
- Facilitated workshops for Indian teachers on critical action pedagogy and curriculum design.

4. PwD Website – Instructional System Design **2023**
Supervisors: Prof. Sridhar Iyer & Prof. Syaamantak Das *IIT Bombay*

- Developed a website and instructional content to raise awareness among higher education functionaries about the needs of persons with disabilities. *[link]*

5. Online Inclusive Library for Deaf and Hard of Hearing **2021**
Supervisor: Prof. Chandan Dasgupta *IIT Bombay*

- Developed a Figma prototype of an online digital library for Deaf and Hard of Hearing learners.
- Created additional curricular resources including a picture book and home sign videos, integrated as core components of the application prototype for scientific concepts.
- Conducted a study to evaluate the effectiveness of the prototype with target users.

Awards and Recognition

• **ISLS Doctoral Consortium Award**, International Society of the Learning Sciences — $2,000 for participation in the 2024 ISLS Doctoral Consortium. **2024**

• **ISAM Travel Scholarship**, International Symposium on Academic Makerspaces — $500 travel scholarship, Sheffield, UK. **2024**

• **Prof. Ram Kumar Scholarship**, awarded by the Prof. Ram Kumar Memorial Foundation at EDM 2023 — $500. **2023**

• **EDM Travel Scholarship**, Educational Data Mining Society — $1,250 travel scholarship. **2023**

Mentoring/Supervision

Intern	**Disha Naurange *(IISER Pune)*** Qualitative research methods focusing on verbal and embodied interaction in collaborative makerspaces.	**2026**
B.Tech	**Oviya *(IIT Bombay)*** Affective computing in collaborative learning settings (supervised research exposition).	**2026**
Research Buddy	**Priya** Academic paper writing as part of PRISm (Practitioner Research Idea Summit), T4E 2025.	**2025**
Research Buddy	**Ganesh Lokhande** Workshop on Documenting Educational Technology Research, T4E 2023	**2023**

Teaching Experience

1. Designing AI-driven Personalised and Adaptive Learning Environments (NPTEL) *[link]* **IIT Bombay, 2026**

- Contributed to content development, instructional design, and course planning.
- Designed assignments and learning activities related to AI-supported personalised learning.
- Recorded demonstration lectures and supported course delivery. *[1] [2] [3] [4]*

2. Learning Analytics and Tools (NPTEL) **IIT Bombay, 2023**

- Designed assignments and practical activities related to learning analytics.
- Mentored learners and facilitated discussion forums.

3. Learning Analytics and Tools (Graduate Course) **IIT Bombay, 2023**

- Supported course delivery, student projects, assignments, and classroom discussions.

4. Human Computer Interaction (Graduate Course) **IIT Bombay, 2022**

- Supported course delivery, student projects, assignments, and classroom discussions.

Skills

Research Methods	Interaction analysis, qualitative coding, interview analysis, affective computing, process analysis, epistemic network analysis (ENA)
Programming	Python, C, C++, MATLAB, HTML, CSS
Tools	ELAN, Figma, Tableau

Invited Talks

1. **Makerspaces as Learning Environments:** Pedagogy and design possibilities for education. *MA Educational Technology, TISS*, July 29, 2026.
2. **TLC Talk Series:** A glimpse into research at the Centre for Educational Technology. *IIT Bombay*, November 8, 2025. *[recording]*

Publications

Journal Articles

1. Garg, A., **Soodhani, N.**, & Rajendran, R. (2025). "Enhancing data analysis and programming skills through structured prompt training: The impact of generative AI in engineering education". *Computers and Education: Artificial Intelligence*, 8, 100380. *[link]*

Journal Articles Under Review

1. **Soodhani, N.**, Rajendran, R., & Iyer, S. "Imitation, Modification, or Innovation? How Task Design Shapes Collaborative Epistemic Practices in Maker Education". Under review at *International Journal of Computer Supported Collaborative Learning*.
2. **Soodhani, N.**, Patil, P., Rajendran, R., & Iyer, S. "Understanding Persistence as Interactional Process: Trouble, Repair, and Epistemic Progress in Collaborative Electronic Making". Under review at *Journal of the Learning Sciences*

Conference Papers

1. **Soodhani, N.**, Rai, A., Rajendran, R., & Iyer, S. (2026). "Characterising Persistence in Collaborative Electronic Making". *ICCE 2026* [Selected].
2. Nishane, I., Badhe, V., Shaha, J., **Soodhani, N.**, Rajmane, A., Rajendran, R., & Iyer, S. (2026). "From Information Foraging to Making Sense of the Information: Decoding Mindset-Driven Learning in Computer Based Learning Environments". *27th International Conference on AI in Education (AIED 2026)*.
3. **Soodhani, N.**, Rajendran, R., & Iyer, S. (2025). "Persistence in Electronic-Making Projects: A Socio-Cognitive Perspective". *International Conference on Computers in Education (ICCE 2025)*.
4. **Soodhani, N.**, Rajendran, R., & Iyer, S. (2025). "Understanding Persistence in Makerspaces: A Socio-Cognitive Perspective on Trouble and Repair Processes". *Proceedings of the 19th International Conference of the Learning Sciences (ICLS 2025)*, pp. 3144–3146. *[link]*
5. **Soodhani, N.**, Prakash, A., Singh, D., Pathan, R., Mishra, A., Shelar, S., Sharma, A., & Rajendran, R. (2024). "Assessing the Dynamics of Middle School Judgment of Learning (JOL) in Mathematics: A Study in Kenya". *International Association for Development of the Information Society (IADIS 2024)*.

6. **Soodhani, N.**, Alekh, V., Saxena, A., & Rajendran, R. (2024). "Unraveling Learner Self-Efficacy in the Electronic Making: Insights from Interaction Analysis, Self-Reports and Interview". *Proceedings of the 18th International Conference of the Learning Sciences (ICLS 2024)*, pp. 1454–1457. *[link]*
7. **Soodhani, N.** (2023). "Analyzing Team Cognition and Combined Efficacy in Makerspaces Using Multimodal Data". *Proceedings of the 16th International Conference on Educational Data Mining (EDM 2023)*, pp. 539–542. *[link]*
8. **Soodhani, N.**, Raman, P., Carvalho, R., Vinay, K., Dasgupta, C., & Slotta, J. D. (2023). "Exploring Teacher Beliefs Around Critical Action Curriculum: Outcomes From a Learning Exchange". *Proceedings of the 17th International Conference of the Learning Sciences (ICLS 2023)*, pp. 2165–2166. *[link]*
9. Carvalho, R., Ghasempour, E., Khan, R., Slotta, J. D., Raman, P., Zhang, X., ..., Dasgupta, C., & **Soodhani, N.** (2023). "Teacher Professional Development in Critical Action Pedagogy: A Culturally Responsive Approach". *Proceedings of the 17th International Conference of the Learning Sciences (ICLS 2023)*, pp. 760–767. *[link]*
10. **Soodhani, N.**, Pawar, M., Kolap, J., Priyadarshini, R., & Dasgupta, C. (2022). "Online Library for Deaf and Hard of Hearing Students". *2022 International Conference on Advanced Learning Technologies (ICALT)*, Bucharest, Romania, pp. 284–286. *[link]*

Poster Presentations

1. **Soodhani, N.**, & Rajendran, R. (2024). "Exploring Academic Makerspace Program for Middle School Learners in Indian Government Schools". *8th International Symposium on Academic Makerspaces (ISAM 2024)*, Sheffield, UK.

Academic Leadership and Community Building

1. **General Secretary, Centre for Educational Technology, IIT Bombay:** Served as the department representative, coordinating academic and student activities. Acted as a liaison between students and faculty. (2023–2024)
2. **Asia-Pacific Representative, International Learning Sciences Student Association (ILSSA):** Organised webinar series: *"ILSSA Odyssey: Exploring Learning Sciences in the Asia-Pacific"* focusing on researchers from the Asia-Pacific region. Contributed to student engagement, networking, and discussions on learning sciences at an international level. (2023–2024)
3. **IRSCP Department Coordinator, IIT Bombay:** Institute Research Scholar Companion Programme - Department Coordinator. (2022–2023)
4. **Conference Service:**
   - PC member for ISLS 2024, T4E 2025, ISLS 2025
   - Sub-reviewer for AIED 2023, ICCE 2023, T4E 2023
   - Volunteer for ISLS 2025, T4E 2023, ICCE 2025, T4E 2025

Educational Outreach

1. **Summer Electronics Workshop:** Designed and conducted a one-month workshop introducing circuits, Arduino, Tinkercad, and block-based programming. Participants engaged in hands-on making with breadboards, LEDs, sensors, and actuators. St. Mary's Malankara High School, Mumbai. (April 2024)
2. **WiSE (Women in Science and Engineering):** Volunteered for hands-on electronics sessions (Light Bulb to Radio, Sense & Act, WiSE Rover) for students from rural India. (May 2024)
3. **Electronic Making Workshop:** Led a two-day workshop on electronic making using breadboards, paper circuits, and electronic textiles for middle school students. (July 2023)

Languages

Tamil (Native) • **English** (Full proficiency) • **Hindi** (Elementary)

References

1. Prof. Ramkumar Rajendran
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2. Prof. Jim Slotta
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3. Prof. Sridhar Iyer
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